Q1 - Integration By Substitution

$I=\int \frac{dx}{e^{x}+4e^{-x}}=f(x)+c$ then $f(x)$ is equal to

(1) $2\tan^{-1}(2e^{x})$

(2) $\frac{1}{2}\tan^{-1}\left(\frac{\mathrm{e}^{\mathrm{x}}}{2}\right)$

(3) $2\tan^{-1}\frac{\mathrm{e}^{\mathrm{x}}}{2}$

(4) $\frac{1}{2}\tan^{-1}\left(2e^{2x}\right)$

Q2 - Integration By Substitution

$I=\int \frac{(1+x)}{x\left(1+xe^{x}\right)^{2}}dx$ is equal to :

(1) $\ln\left(\frac{xe^{x}}{1+xe^{x}}\right)-\left(\frac{1}{1+xe^{x}}\right)+c$

(2) $\ln\left(\frac{xe^{x}}{1+xe^{x}}\right)+\left(\frac{1}{1-xe^{x}}\right)+c$

(3) $\ln\left(\frac{xe^{x}}{1-xe^{x}}\right)+\left(\frac{1}{1+xe^{x}}\right)+c$

(4) $\ln\left(\frac{xe^{x}}{1+xe^{x}}\right)+\left(\frac{1}{1+xe^{x}}\right)+c$

Q3 - Integration By Substitution

$\int \frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx$ is equal to :

(1) $\frac{\sin 2x}{2}-\sin x+c$

(2) $-\frac{\sin 2x}{2}+\sin x+c$

(3) $-\frac{\sin 2x}{2}-\sin x+c$

(4) $\frac{\sin 2x}{2}+\sin x+c$

Q4 - Integration By Substitution

$\int \frac{e^{x}(1+x)}{\cos^{2}\left(xe^{x}\right)}dx=$

(1) $-\cot\left(xe^{x}\right)$

(2) $\tan\left(xe^{x}\right)$

(3) $\tan\left(e^{x}\right)$

(4) none of these

Q5 - Integration By Substitution

$\int \tan^3 2x \sec 2x dx =$

(1) $\sec^3 2x + 3\sec 2x$

(2) $\frac{1}{6}\left[\sec^3 2x - 3\sec 2x\right]$

(3) $\left[\sec^3 2x - 3\sec 2x\right]$

(4) none of these

Q6 - Integration By Substitution

$\int \frac{\log(x+1)-\log x}{x(x+1)} dx$ equals

(1) $-\log\left(\frac{x+1}{x}\right) + c$

(2) $-\log\left[\log\left(\frac{x+1}{x}\right)\right] + c$

(3) $-\frac{1}{2}\left[\log\left(\frac{x+1}{x}\right)\right]^2 + c$

(4) $c - \frac{1}{2}\left[\log(x+1)^2 - (\log)^2\right]$

Q7 - Integration By Substitution

$\int \frac{\sin x - \cos x}{\sqrt{1-\sin 2x}} e^{\sin x} \cos x dx$ is equal to

(1) $e^{\sin x} + c$

(2) $e^{\sin x - \cos x} + c$

(3) $e^{\sin x + \cos x} + c$

(4) $e^{\cos x - \sin x} + c$

Q8 - Integration By Substitution

$\int \frac{\cos 2x}{\cos x} dx$ is equal to

(1) $2\sin x + \log(\sec x - \tan x) + c$

(2) $2 \sin x-\log (\sec x-\tan x)+c$

(3) $2 \sin x+\log (\sec x+\tan x)+c$

(4) $2 \sin x-\log (\sec x+\tan x)+c$

Q9 - Integration By Substitution

$\int e^{\log _5 x} d x$

(1) $\frac{x^{\log _5 e}}{\log _5 e}$

(2) $\frac{x^{\log _5 5 e}}{\log _5 5 e}$

(3) $\frac{x^{\log _e 5 e}+1}{\log _e 5 e+1}$

(4) none of these

Q10 - Integration By Substitution

$\int\left\{\frac{(1-\cos x)}{[\cos x(1+\cos x)]}\right\} d x=$

(1) $\log (\sec x+\tan x)-2 \tan \left(\frac{x}{2}\right)$

(2) $\log (\sec x+\tan x)+2 \tan \left(\frac{x}{2}\right)$

(3) $\log (\sec) \tan \left(\frac{x}{2}\right)$

(4) none of these

Q11 - Integration By Substitution

$\int(1-x)^{23} x d x$

(1) $\frac{x^{23}}{23}+\frac{x^{24}}{24}+c$

(2) $\frac{(x-1)^{23}}{23}+\frac{(x-1)^{24}}{24}+c$

(3) $\frac{(1-x)^{25}}{25}-\frac{(1-x)^{24}}{24}+c$

(4) none of these

Q12 - Integration By Substitution

$\int \frac{d x}{\cos ^6 x+\sin ^6 x}$ is equal to

(1) $\log_e(\tan x-\cot x)+c$

(2) $\log_e(\cot x-\tan x)+c$

(3) $\tan^{-1}(\tan x-\cot x)+c$

(4) $\tan^{-1}(2\cot 2x)+c$

Q13 - Integration By Substitution

$\int\left(e^{\log x}+\sin x\right)\cos x dx$ is equal to

(1) $x\sin x+\cos x-\sin^2 x+c$

(2) $x\cos x-\sin^2 x+c$

(3) $x\sin x+\cos x-\left(\cos^2 x\right)/2+c$

(4) $x^2\sin x+\cos x-\sin^3 x+c$

Q14 - Integration By Substitution

$\int\frac{x+\sin x}{1+\cos x}dx$ is equal to

(1) $x\tan\frac{x}{2}+c$

(2) $\tan\frac{x}{2}+c$

(3) $x\tan x+c$

(4) $\frac{x}{2}\tan\frac{x}{2}+c$

Q15 - Integration By Substitution

$\int x^x(1+\log x)dx$ is equal to

(1) x^x-c

(2) x^x+c

(3) $x\log x+c$

(4) None of these

Q16 - Integration By Substitution

$\int \frac{\sin x+\cos x}{9+16 \sin 2x} dx$ is equal to :

(1) $\frac{1}{40} \ln \left|\frac{5+4(\sin x-\cos x)}{5-4(\sin x-\cos x)}\right|+c$

(2) $\frac{1}{40} \ln \left|\frac{5-4(\sin x-\cos x)}{5+4(\sin x-\cos x)}\right|+c$

(3) $\frac{1}{40} \ln \left|\frac{5+4(\sin x+\cos x)}{5-4(\sin x+\cos x)}\right|+c$

(4) $\frac{1}{40} \ln \left|\frac{5-4(\sin x+\cos x)}{5+4(\sin x-\cos x)}\right|+c$

Q17 - Integration By Substitution

$\int \frac{\left(\sqrt{x^2+1}\right)\left(\ln\left(x^2+1\right)-2 \ln x\right\}}{x^4} dx$ is equal to

(1) $\frac{\left(x^2+1\right)\sqrt{x^2+1}}{x^3}\left[2-3 \ln \left(\frac{x^2+1}{x^2}\right)\right]+c$

(2) $\frac{1}{9} \frac{\left(x^2+1\right)\sqrt{x^2+1}}{x^3}\left[2+3 \ln \left(\frac{x^2+1}{x^2}\right)\right]+c$

(3) $\frac{\left(x^2+1\right)\sqrt{x^2+1}}{x^3}\left[2+3 \ln \left(\frac{x^2+1}{x^2}\right)\right]+c$

(4) $\frac{1}{9} \frac{\left(x^2+1\right)\sqrt{x^2+1}}{x^3}\left[2-3 \ln \left(\frac{x^2+1}{x^2}\right)\right]+c$

Q18 - Integration By Substitution

$\int \frac{\cos 2x}{\cos x+\sin x} dx$ is equal to

(1) $\sin x-\cos x+c$

(2) $-\sin x+\cos x+c$

(3) $\sin x+\cos x+c$

(4) none of these

Q19 - Integration By Substitution

$\int \frac{\mathrm{e}^{\mathrm{x}-1}+\mathrm{x}^{\mathrm{e}-1}}{\mathrm{e}^{\mathrm{x}}+\mathrm{x}^{\mathrm{e}}} \mathrm{dx}$ is equal to

(1) $\frac{1}{e} \ln \left(x^e+e^x\right)+c$

(2) $\frac{1}{e} \ln (x+e)+c$

(3) $\frac{1}{e} \ln \left(x^{-e}+e^{-x}\right)+c$

(4) none of these

Q20 - Integration By Parts

If $\int x e^{x} \cos x d x .=f(x)+c$, then $f(x)$ is equal to

(1) $\frac{e^{x}}{2}\{(1-x) \sin x-x \cos x\}$

(2) $\frac{\mathrm{e}^{\mathrm{x}}}{2}\{(1-\mathrm{x}) \sin \mathrm{x}+\mathrm{x} \cos \mathrm{x}\}$

(3) $\frac{\mathrm{e}^{\mathrm{x}}}{2}\{(1+\mathrm{x}) \sin \mathrm{x}-\mathrm{x} \cos \mathrm{x}\}$

(4) none of these

Q21 - Integration By Parts

$\int x^{2} \sin x d x$

(1) $x^{2} \sin x-2 x \cos x+c$

(2) $x^{2} \sin x+c$

(3) $-x^{2} \cos x+2 x \sin x+2 \cos x+c$

(4) $-x^{2} \sin x-2 x \cos x+\sin x+c$

Q22 - Integration By Parts

$\int \frac{x \cos x}{(x \sin x+\cos x)^{2}} d x$ is equal to

(1) $\frac{1}{(x \sin x+\cos x)^{2}}+k$

(2) $-\frac{1}{(x \sin x+\cos x)}+k$

(3) $\frac{1}{(x \sin x+\cos x)^{3}}+k$

(4) $\frac{1}{(x \sin x+\cos x)^{4}}+k$

Q23 - Integration By Parts

$\int e^{\tan ^{-1} x}\left(\frac{1+x+x^{2}}{1+x^{2}}\right) d x$ is equal to

(1) $x e^{\tan ^{-1} x}+c$

(2) $x^{2} e^{\tan ^{-1} x}+c$

(3) $\frac{1}{x e^{\tan ^{-1} x}}+c$

(4) $\frac{1}{x^2} e^{\tan^{-1} x}+c$

Q24 - Integration By Parts

If $\int \frac{\sqrt{\cot x}}{\sin x \cos x} d x=A \sqrt{\cot x}+B$, then A is equal to

(1) 1

(2) 2

(3) -1

(4) -2

Q25 - Integration By Parts

The value of $\int \frac{\log (x / e)}{(\log x)^2} d x$ is :

(1) $\frac{x+1}{(\log x)^2}+c$

(2) $\frac{x-1}{(\log x)^2}+c$

(3) $\frac{x}{\log x}+c$

(4) $\frac{\log x}{x}+c$

Q26 - Integration By Parts

The value of the integral $\int \mathrm{e}^{\sin ^2 x}\left(\cos x+\cos ^3 x\right) \sin x d x$ is

(1) $\frac{1}{2} e^{\sin ^2 x}\left(3-\sin ^2 x\right)+c$

(2) $e^{\sin ^2 x}\left(1+\frac{1}{2} \cos ^2 x\right)+c$

(3) $e^{\sin ^2 x}\left(3 \cos ^2 x+2 \sin ^2 x\right)+c$

(4) $e^{\sin ^2 x}\left(2 \cos ^2 x+3 \sin ^2 x\right)+c$

Q27 - Special Substitutions and Integrals

$\int \frac{d x}{\sqrt{(x-a)(b-x)}}=$

(1) $2 \sin ^{-1} \sqrt{\left(\frac{x-a}{b-a}\right)}+c$

(2) $\sin^{-1}\sqrt{\left(\frac{x-a}{b-a}\right)}+c$

(3) $2\sin^{-1}\sqrt{\left(\frac{x+a}{b-a}\right)}+c$

(4) none of these

Q28 - Special Substitutions and Integrals

$\int\frac{\sin x}{\sin x-\cos x}dx$ is equal to :

(1) $\frac{x}{2}-\frac{1}{2}\log\sin x-\cos x+c$

(2) $\frac{x}{2}+\frac{1}{2}\log\sin x-\cos x+c$

(3) $\frac{x}{2}+\frac{1}{2}\log\sin x+\cos x+c$

(4) none of these

Q29 - Special Substitutions and Integrals

$\int\frac{(x^4-x)^{\frac{1}{4}}}{x^5}dx$ is equal to

(1) $\frac{4}{15}\left(1-\frac{1}{x^3}\right)^{\frac{5}{4}}+c$

(2) $\frac{4}{5}\left(1-\frac{1}{x^3}\right)^{\frac{5}{4}}+c$

(3) $\frac{4}{15}\left(1+\frac{1}{x^3}\right)^{\frac{5}{4}}+c$

(4) none of these

Q30 - Special Substitutions and Integrals

$\int\frac{3+2\cos x}{(2+3\cos x)^2}dx$ is equal to

(1) $\left(\frac{\sin x}{3\cos x+2}\right)+c$

(2) $\left(\frac{2\cos x}{3\sin x+2}\right)+c$

(3) $\left(\frac{2\cos x}{3\cos x+2}\right)+c$

(4) $\left(\frac{2\sin x}{3\sin x+2}\right)+c$

Q31 - Special Substitutions and Integrals

$\int \frac{dx}{(2x+1)(1+\sqrt{(2x+1)}}$ is equal to :

(1) $\tan^{-1} \frac{\sqrt{2x+1}}{1+\sqrt{2x+1}} + c$

(2) $\log_e \frac{\sqrt{2x+1}}{1+\sqrt{2x+1}} + c$

(3) $\log_e\left(\frac{1+\sqrt{2x+1}}{\sqrt{2x+1}}\right) + c$

(4) $\tan^{-1} \frac{1+\sqrt{2x+1}}{\sqrt{2x+1}} + c$

Q32 - Special Substitutions and Integrals

$I = \int \frac{\left(x+x^{\frac{2}{3}}+x^{\frac{1}{6}}\right)}{x\left(1+x^{\frac{1}{3}}\right)} dx$ is equal to

(1) $\frac{3}{2}x^{\frac{2}{3}} + 6\tan^{-1}\left(x^{\frac{1}{6}}\right) + c$

(2) $\frac{3}{2}x^{\frac{2}{3}} - 6\tan^{-1}\left(x^{\frac{1}{6}}\right) + c$

(3) $\frac{3}{2}\mathrm{x}^{\frac{2}{3}} + \tan^{-1}\left(\mathrm{x}^{\frac{1}{6}}\right) + \mathrm{C}$

(4) none of these

Q33 - Special Substitutions and Integrals

$\int \sqrt{\tan x} + \sqrt{\cot x} dx$ is equal to :

(1) $\sqrt{2}\sin^{-1}(\sin x + \cos x) + c$

(2) $\frac{1}{\sqrt{2}}\sin^{-1}(\sin x - \cos x) + c$

(3) $\sqrt{2}\sin^{-1}(\sin x - \cos x) + c$

(4) $\frac{1}{\sqrt{2}}\sin^{-1}(\sin x + \cos x) + c$

Q34 - Special Substitutions and Integrals

$\int \frac{1}{x^2(x^4+1)^{3/4}} dx$ is equal to :

(1) $\left(1+\frac{1}{x^4}\right)^{1/4} + c$

(2) $-\left(1+\frac{1}{x^4}\right)^{1/4} + c$

(3) $-\left(1-\frac{1}{x^4}\right)^{1/4}+c$

(4) $\left(1-\frac{1}{x^4}\right)^{1/4}+c$

Q35 - Special Substitutions and Integrals

$\int \frac{\sqrt{5+x^{10}}}{x^{16}} dx$ is equal to :

(1) $-\frac{1}{75}\left(1+\frac{5}{x^{10}}\right)^{3/2}+c$

(2) $-\frac{1}{75}\left(1-\frac{5}{x^{10}}\right)^{3/2}+c$

(3) $\frac{1}{75}\left(1-\frac{5}{x^{10}}\right)^{3/2}+c$

(4) $\frac{1}{75}\left(1+\frac{5}{x^{10}}\right)^{3/2}+c$

Q36 - Special Substitutions and Integrals

$\int \frac{dx}{5+4\cos x}$ is equal to :

(1) $\frac{2}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right)+c$

(2) $-\frac{2}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right)+c$

(3) $\frac{1}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right)+c$

(4) None of these

Q37 - Special Substitutions and Integrals

$\int \frac{dx}{(2x+3)\sqrt{4x+5}}$ is equal to :

(1) $\tan^{-1}\sqrt{4x-5}+c$

(2) $\tan^{-1}\sqrt{4x+5}+c$

(3) $\tan^{-1}\sqrt{5x+4}+c$

(4) $\tan^{-1}\sqrt{5x-4}+c$

Q38 - Special Substitutions and Integrals

$\int e^{ax}\cos bx\, dx$ is equal to

(1) real part of $\int e^{(a+bi)x}dx$

(2) imaginary part of $\int \mathrm{e}^{(\mathrm{a}+\mathrm{bi})\mathrm{x}}\mathrm{dx}$

(3) neither real nor Imaginary part of $\int \mathrm{e}^{(a+bi)x}dx$

(4) none of these

Q39 - Special Substitutions and Integrals

If $I_n=\int(\ln x)^n dx$ then I_n+nl_{n-1} is equal to

(1) $x(\ln x)^{n-1}$

(2) $x(\ln x)^n$

(3) $\mathrm{nx}(\ln \mathrm{x})^{\mathrm{n}}$

(4) none of these

Q40 - Special Substitutions and Integrals

$\int \frac{dx}{x-x^3}=A\log\left|\frac{x^2}{1-x^2}\right|+c$ then A is equal to

(1) 2

(2) $\frac{1}{2}$

(3) $\frac{2}{3}$

(4) $\frac{1}{3}$

Answer Key

Q1 (2)	**Q2** (4)	**Q3** (3)	**Q4** (2)
Q5 (2)	**Q6** (3)	**Q7** (1)	**Q8** (4)
Q9 (2)	**Q10** (1)	**Q11** (3)	**Q12** (3)
Q13 (3)	**Q14** (1)	**Q15** (2)	**Q16** (1)
Q17 (4)	**Q18** (3)	**Q19** (1)	**Q20** (1)
Q21 (3)	**Q22** (2)	**Q23** (1)	**Q24** (4)
Q25 (3)	**Q26** (1, 2)	**Q27** (1)	**Q28** (2)
Q29 (1)	**Q30** (1)	**Q31** (1)	**Q32** (1)
Q33 (3)	**Q34** (2)	**Q35** (1)	**Q36** (1)
Q37 (2)	**Q38** (1)	**Q39** (2)	**Q40** (2)